

Motorcycle Datalogger & Visualization

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Project Overview

1.1 What is the Problem?

Motorcycles are frequently ridden aggressively in both street and track environments, but a major distinction between everyday riders and professional racers (Besides the absolutely immense skill gap) is the ability to obtain quantitative post-ride feedback. Quantitative telemetry, such as braking behavior, acceleration, and lean dynamics, allows professionals to analyze and improve their riding from a more objective perspective than the typical rider has access to with their more subjective perception of what is going on around them. To address this gap, several commercial-level motorcycle data-logging systems have been created. These systems generally fall into two main categories. The first consists of fully proprietary systems with dedicated sensors, computers, and wiring, but these systems are often quite expensive, frequently running up a bill for thousands of dollars. The second category is more affordable and piggybacks on existing CAN bus data from factory-installed sensors. However, this approach is only viable for modern motorcycles equipped with modern electronics and sensor suites. Older motorcycles, particularly those more than a decade old, lack these sensors entirely. For example, the 2001 Suzuki SV650 (my motorcycle) completely lacks traction control, ABS, and lean-angle sensing, making it incompatible with modern telemetry solutions that piggyback on the CAN. With this in mind it becomes clear why those with older vehicles are somewhat out of options if they don't want to spend multiple thousands of dollars, but the bright side is that it creates a technical opportunity to develop a self-contained, low-cost system capable of collecting meaningful riding data independently of factory electronics, while remaining robust in the electrically noisy and vibration-prone environment of a motorcycle. The purpose of the project is to demonstrate the feasibility of a low-cost, modular CAN-based embedded system capable of reliably acquiring and analyzing motorcycle kinematic data using independent sensors, enabling accurate and quantifiable rider feedback on braking, acceleration, and lean angle behavior to give a more objective perspective on rider skills without relying on factory electronics beyond a basic power input.

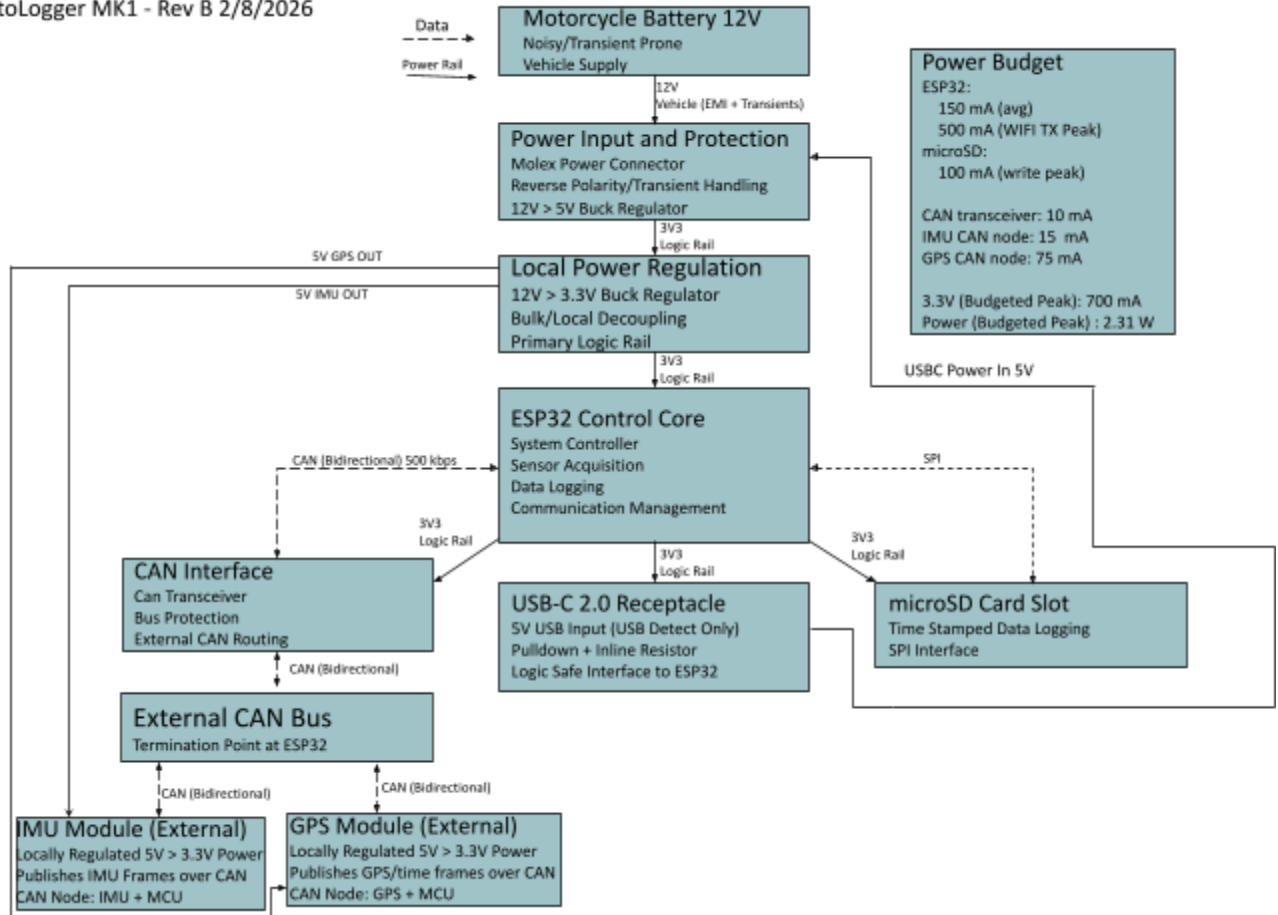
1.2 System Overview

The system is designed as a modular embedded platform capable of acquiring, transmitting, and processing motorcycle kinematic data in real time. It consists of a motherboard responsible for the main processing and real-time computation of collected data, distributed daughterboard sensor modules, a protected and conditioned power input stage, and a Controller Area Network (CAN) communication system that interconnects all sensor nodes with the motherboard. The architecture is designed for robust power cleaning, modular expansion, and reliable operation in the high-noise, high-heat, high-vibration motorcycle environment. The core of the system is the central motherboard, which performs primary processing, data storage, and CAN coordination. The motherboard receives power from the motorcycle's electrical system via the onboard protected power input that mitigates reverse polarity, voltage transients, and electrical noise commonly produced by stator-driven charging systems. Following protection and regulation, clean power is distributed to onboard circuitry and external daughterboard modules while maintaining electrical stability and minimizing noise propagation throughout the system. The sensor data will be collected from independent modular sensor daughterboards rather than relying on factory-installed electronics (My bike has none anyway). To make it easier to create an initial prototype, only two sensor daughterboards will be installed initially. The first will be an Inertial Measurement Unit (IMU), and the second will be a GPS system. The purpose of the IMU is to measure angular rate and acceleration data from accelerometers and gyroscopes, enabling an accurate estimate of orientation and dynamic motion. The GPS will provide velocity and positional reference data, which can support kinematic analysis and be used on its own for mapping/visualization. The sensors will communicate with the motherboard over CAN, which was chosen for its differential signaling, noise resistance, reliability in electrically harsh environments, and high adoption in the automotive industry. The entire system operates by the sensors continuously sending data onto the CAN bus for the motherboard to receive. The ESP32 microcontroller on the motherboard will handle initial data processing and prepare to send the data to the microSD card, where it will be stored for post-ride analysis. The collected data can be used to learn more about rider

behavior, such as acceleration characteristics, lean-angle trends, or braking behavior. The data can be visualized later to show the quantitative rider feedback. As of right now, there will be no real-time calculations besides sensor fusion and data filtering. The actual real-time kinematic calculations will come after the first revision of the prototype has been built and functions correctly. The modular architecture of the entire system enables the user (Me) to add sensing or processing nodes without a major redesign of core systems or daughterboards. The combination of a protected automotive power design, robust CAN communication, and independent sensing means the system is an entirely self-contained platform capable of outputting meaningful, quantifiable motorcycle performance data without relying on anything from the motorcycle other than a power input.

Below is a visual block diagram of the overarching system architecture to help contextualize the above paragraph. Dashed lines are DATA only lines and solid lines are POWER only lines. To the right is a rough power budget for high-current loads from the main components, to support rough simulation/verification of the power input stage in PSPICE in Cadence ORCAD.

MotoLogger MK1 - Rev B 2/8/2026



1.3 Engineering Challenges

Preface

I must preface the next section by saying that this is a constantly updated list, as I have not yet finished this project. There will be mistakes and problems along the way, due to both the project's complexity and the fact that I'm working on it in my spare time by myself, learning as I go.

Electrical Noise and Transient Conditions

Automotive charging systems in cars and motorcycles generate much more noise and voltage transients than a bench variable power supply. More specifically, on a motorcycle, they generally have an engine-driven stator connected to a shunt-type regulator/rectifier that

converts AC to DC and lowers it to 12ish volts. The problem is that they *also* generate voltage transient spikes and significant electrical noise. Reverse polarity events, load dumps, and high-frequency switching noise from the motherboards' on-board buck regulators can disrupt logic operation or damage sensitive electronics. To address power input issues, the system incorporates a protected power input stage that mitigates reverse polarity events, suppresses transient voltage spikes, and cleans the incoming power before onboard regulation and distribution.

Post-Regulation Power Stability Under Dynamic Load

The entire system (as of right now) will run on 3.3V, which means there will be a voltage regulator to drop the 12V down to 5V, then to 3.3V. That doesn't matter much when the PCB isn't on a motorcycle, where it will have much more stable power, but when on a motorcycle and 12-14 volt power isn't guaranteed for every second the device is powered on, there must be thought put into brownout prevention and high frequency noise/ripple removal so that the power entering the motherboard components and entering the daughterboards is as clean as possible. The solution to most of these issues is to add the necessary electrolytic caps for low-frequency ripple suppression and ceramic caps for high-frequency noise suppression. As an aside, I feel it's apt to mention that under starter cranking, voltage drops quite low, possibly to 7 or 8 volts. While I don't plan to have the device on while the bike is starting, I have chosen switching regulators with a minimum input voltage of 7V, so it should still function during most cranking, unless the battery is old.

Reliable CAN Communication in a High EMI, High Vibration Environment

Motorcycles are hotbeds for high EMI and vibration due to the nature of being a metal frame strapped to an engine with two wheels. Even though CAN is known for its robustness in automotive applications, that doesn't mean that there can't be intermittent connections and spotty data transmission if the environment sucks and no precautions were taken. This isn't to say it's useless, but that there must be some preparation done in case the differential signaling, noise resistance, and known reliability aren't enough. This means including pads for common-mode chokes if common-mode noise becomes prevalent, ensuring the traces on all

boards are as symmetrical and equal as possible, and using locking connectors to reduce the risk of loose connections.

Thermal and Mechanical Stress

Because the project is still in the PCB design verification and development phase, little thought has been given to how exactly the boards will be mounted to mitigate aggressive vibration and heat. A concrete strategy will be established once the PCB prototype has been bench tested. That said, the SV650 features a 90° V-Twin engine layout, which is a plus for this project. Normally, V-Twins aren't balanced at all and are known for aggressive vibrations (Harleys). The 90° V-Twin is a special type of V-Twin because it has primary balance, meaning the pistons' vibrations cancel each other out and are less vibrationally aggressive than in some other engine layouts. That does *not* mean it doesn't have vibrations; however, the engine still isn't balanced on the secondary, and it's still an *engine*, so the vibration removal can only go so far. That means the casings holding the PCBs for this must have some level of vibration mitigation, and the board's physical connectors must have locking mechanisms to prevent loose connections and flaky signals. The heat problem is easier to mitigate as long as the boards aren't placed near the engine. There is ample room behind the rider by the battery where these boards can go, which doesn't get nearly as hot as the area around the engine.

System Hardware Design

2.1 Project Recap

The system is designed as a modular embedded platform that consists of an ESP32-powered motherboard, daughterboard sensor modules, protected power input, and a CAN-based protocol. The architecture is designed to be reliable in an electrically noisy, mechanically demanding motorcycle environment while maintaining modular expansion capability and stable system performance. The following section describes the hardware implementation of the system, including the power conditioning and regulation architecture, motherboard processing core, communication infrastructure, and sensor integration. Design

decisions were made to prioritize EMI resistance, power stability, and reliable data acquisition and transmission under real-world conditions.

ESP32-S3-WROOM-1U-N16R8

IO Pins used:

Power Pins

3v3 - Cleaned Motorcycle Power In

GND - Common Ground on Both PCB Layers Connected by Ground Vias

EN - Pullup Resistor coming off of 3v3 rail - indicates Power on state to chip

GPIO 0 / Pin 27 - Boot circuit for flashing firmware connected to 3v3 via pullup resistor with a momentary switch to ground

CANBUS Transceiver GPIO

GPIO 38 / Pin 31 - CAN Transmit Data IN - IC Pin D

GPIO 21 / Pin 23 - CAN Receive Data OUT - IC Pin R

microSD Card GPIO - SPI Mode

GPIO 10 / Pin 18 - Chip Select normally pulled high to 3v3, but can be overridden by ESP32-DAT3

GPIO 11 / Pin 19 - SPI_MOSI ESP32 write to microSD card-CMD

GPIO 12 / Pin 20 - Clock Sync driven by ESP32-CLK

GPIO 13 / Pin 21 - SPI_MISO SD card to ESP32-DAT0

USB-C 2.0 GPIO

GPIO 4 / Pin 4 - USB Detect Pin

GPIO 19 / USB_D- / Pin 13 - USB Differential Negative

GPIO 20 / USB_D+ / Pin 14 - USB Differential Positive

LED Indicator GPIO

GPIO 14 / Pin 22 - Firmware Indicator

10 Pin Connector GPIO

GPIO 35 / Pin 28 - Log Arm Switch

GPIO 36 / Pin 29 - CAN Enable Switch

GPIO 37 / Pin 30 - Log Green LED

GPIO 38 / Pin 31 - Log Red LED

GPIO 39 / Pin 32 - CAN Green LED

GPIO 40 / Pin 33 - CAN Red LED

GPIO 41 / Pin 34 - GPS Active Indicator LED

GPIO 46 / Pin 16 - Event Flagger (Momentary Button/Switch)

Switches Off Board

Power On - Will be on Moto Battery lines and high power switch

CAN Enable - Multi colored LED

Yellow: On, but not actively transmitting

Green: On and Transmitting

Red/Blinks: Error Codes

Activate Logging / Write to SD - Multi Colored LED

Yellow: On, but not writing

Green: On and writing

Red: Error Codes

Motherboard Master Parts Reference List

Diodes

D1_RP - Reverse Polarity Protection Schottky Diode - SS54FSH

D2_TVS - TVS Zener Diode - SMBJ18A

Resistors

R1 - 10k Ohm - MCU EN Pullup

R2 - 10k Ohm - MCU IOBoot Pullup

R3 - 1.8k Ohm - Power Indication LED Resistor

R4 - 5.1k Ohm - USBC CC1 Pulldown

R5 - 5.1k Ohm - USBC CC2 Pulldown

R6 - 10k Ohm - microSD Dat0 Pullup

R7 - 10k Ohm - microSD CMD Pullup

R8 - 120 Ohm - CAN Termination Resistor

R9 - 10k Ohm - Event Flagger Pullup

R10 - 220 Ohm - Firmware Indicator LED Resistor

R11 - 220 Ohm - Event Flagger Connector Resistor

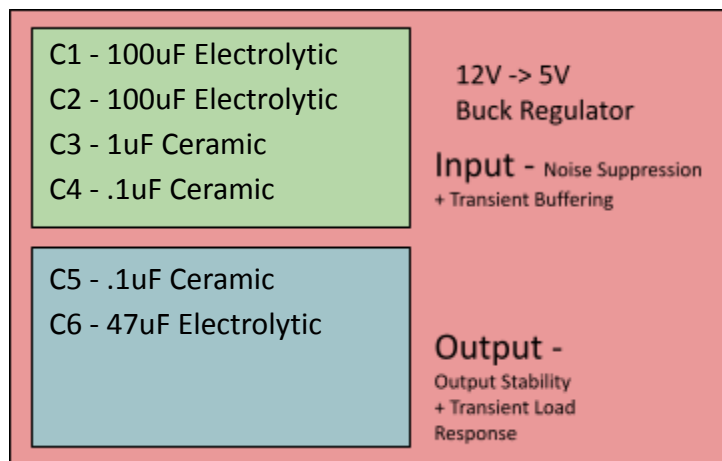
R12 - 10k Ohm - microSD Dat3/CD Pullup

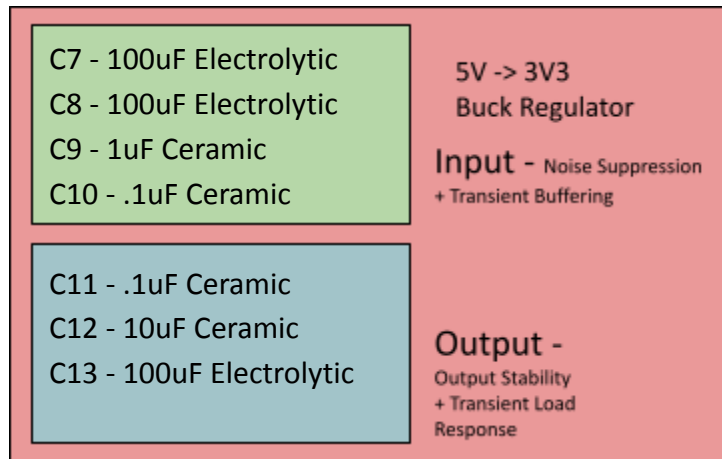
Ferrite Beads

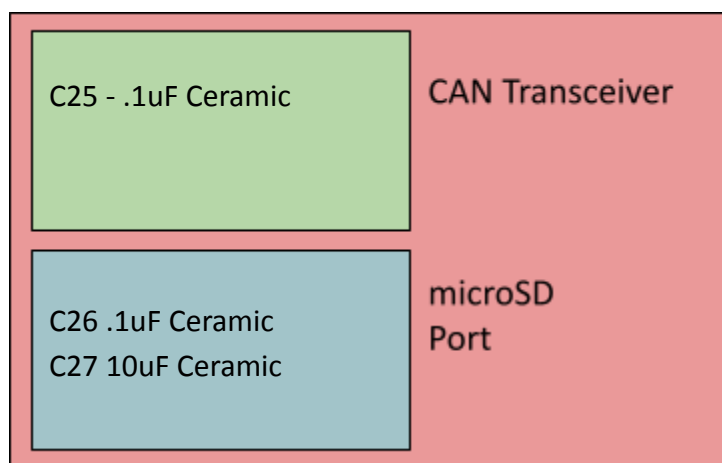
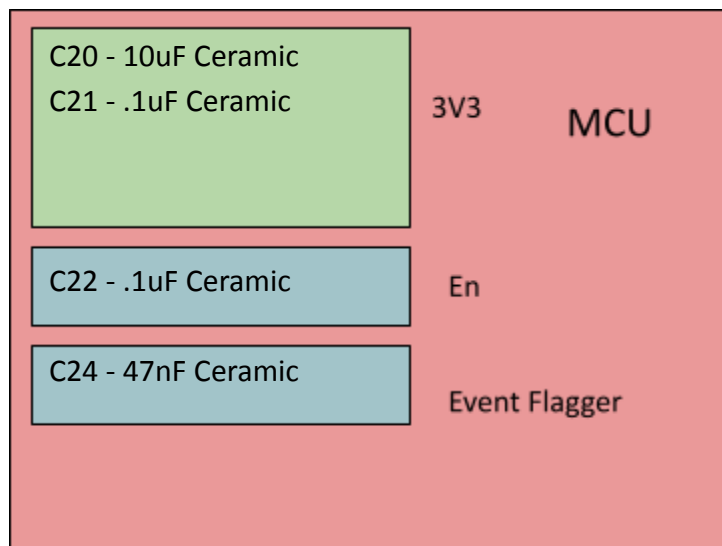
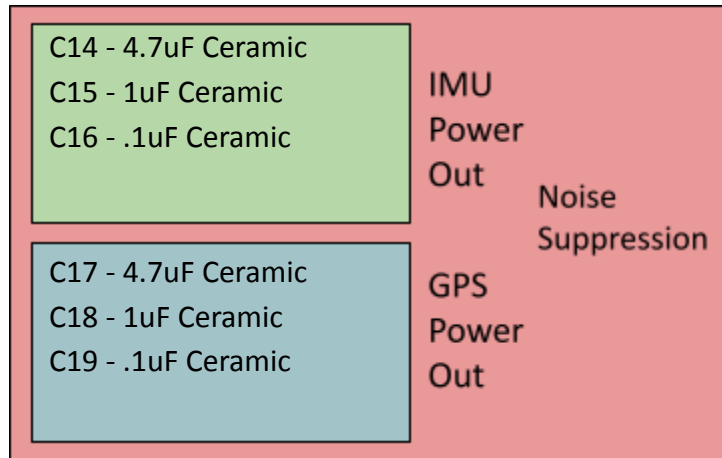
FB1 - IMU Power Out - 742792113

FB2 - GPS Power Out - 742792113

Capacitors







To Do:

~~Create an Architectural Block Diagram~~

Finish going through the motherboard parts and audit all components **(In Progress)**

Finish IMU daughterboard

Finish GPS daughterboard

Simulate and Document power rail testing with SPICE **(In Progress)**

Create a write-up documenting every part used and why it was used (Main components, not resistors, caps, etc.) **(In Progress)**

Move GPIO decoupling caps to the bottom layer and rework top-layer traces

Shorten power rail paths to external power connectors

Rework traces and ground vias to have better ground pours and less breaks

Hardware Design Justification and Verification

SS54FSH - Schottky Diode

A Schottky diode is used for reverse-polarity protection in the event of crossed battery leads. The Schottky diode was chosen over a PN diode due to its lower forward voltage drop and fast switching / no reverse recovery. The SS54FSH was specifically chosen for its automotive-grade performance and forward current of 5A, which is well above the estimated peak current load of 700mA. The device's higher peak amperage enables better thermal performance on the board, providing greater headroom relative to the typical current load expected from the components. The SS54FSH forward voltage varies with current and temperature, typically ranging from 0.35 to 0.65 V. Lower forward voltage reduces power dissipation during normal operation. The SS54FSH has a maximum DC reverse voltage of 40V, providing adequate margin for the expected maximum typical voltage of 16V at the 12V power input.

Power Dissipated

$$P_D = I_F \cdot V_F$$

$$.7A \text{ (Estimated Peak)} \cdot \sim .35V = .245W$$

$$.7A \text{ (Estimated Peak)} \cdot .48V = .336W$$

Power Dissipated Approximately .245-.336 Watts

Absolute Junction Temperature was estimated using the thermal resistance relationship given by:

$$T_J = T_A + P_D \cdot \theta_{JA}$$

T_J = Absolute Junction Temperature

T_A = Ambient Temperature

P_D = Power Dissipated

θ_{JA} = Junction to Ambient Thermal Resistance

$$25^\circ\text{C} + .245 \text{ W} \cdot 63^\circ\text{C/W} \approx 40^\circ\text{C}$$

40°C Junction temperature at 25°C ambient temperature with .245W power dissipation

$$25^\circ\text{C} + .336 \text{ W} \cdot 63^\circ\text{C/W} \approx 46^\circ\text{C}$$

46°C junction temperature at 25°C ambient temperature with .336W power dissipation

The copper pour the diode is sitting on for the front layer is approximately 43mm², which is larger than the 5mm x 5mm copper pad for thermal performance testing. The motologger board also has a 7.2mm x 5.8mm bottom-layer copper pad connected by vias for improved thermal performance, effectively doubling the total area to ~80mm².

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted)							
PARAMETER	SYMBOL	SS54 FSH	SS56 FSH	SS510 FSH	SS515 FSH	SS520 FSH	UNIT
Marking code on the device		54FSH	56FSH	510FSH	515FSH	520FSH	
Repetitive peak reverse voltage	V_{RRM}	40	60	100	150	200	V
Reverse voltage, total rms value	$V_{R(RMS)}$	28	42	70	105	140	V
Forward current	I_F	5					A
Surge peak forward current, single half sine-wave superimposed on rated load	$t = 8.3\text{ms}$	I_{FSM}					A
	$t = 1.0\text{ms}$						A
Junction temperature	T_J	-55 to +150					$^\circ\text{C}$
Storage temperature	T_{STG}	-55 to +150					$^\circ\text{C}$

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-lead thermal resistance	$R_{\theta JA}$	13	$^\circ\text{C/W}$
Junction-to-ambient thermal resistance	$R_{\theta JA}$	63	$^\circ\text{C/W}$
Junction-to-case thermal resistance	$R_{\theta JC}$	10	$^\circ\text{C/W}$

Thermal Performance Note: Units mounted on PCB (5mm x 5mm Cu pad test board)

ELECTRICAL SPECIFICATIONS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

PARAMETER	CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage ⁽¹⁾	SS54FSH $I_F = 2.5\text{A}, T_J = 25^\circ\text{C}$	V_F	0.48	-	V
	SS54FSH $I_F = 5.0\text{A}, T_J = 25^\circ\text{C}$		0.58	0.65	V
	SS54FSH $I_F = 2.5\text{A}, T_J = 125^\circ\text{C}$		0.42	-	V
	SS54FSH $I_F = 5.0\text{A}, T_J = 125^\circ\text{C}$		0.53	0.62	V
	SS56FSH $I_F = 2.5\text{A}, T_J = 25^\circ\text{C}$		0.49	-	V
	SS56FSH $I_F = 5.0\text{A}, T_J = 25^\circ\text{C}$		0.59	0.70	V
	SS56FSH $I_F = 2.5\text{A}, T_J = 125^\circ\text{C}$		0.43	-	V
	SS56FSH $I_F = 5.0\text{A}, T_J = 125^\circ\text{C}$		0.54	0.63	V
	SS510FSH $I_F = 2.5\text{A}, T_J = 25^\circ\text{C}$		0.68	-	V
	SS510FSH $I_F = 5.0\text{A}, T_J = 25^\circ\text{C}$		0.76	0.85	V
	SS510FSH $I_F = 2.5\text{A}, T_J = 125^\circ\text{C}$		0.55	-	V
	SS510FSH $I_F = 5.0\text{A}, T_J = 125^\circ\text{C}$		0.63	0.80	V
	SS515FSH $I_F = 2.5\text{A}, T_J = 25^\circ\text{C}$		0.75	-	V
	SS515FSH $I_F = 5.0\text{A}, T_J = 25^\circ\text{C}$		0.81	0.95	V
	SS515FSH $I_F = 2.5\text{A}, T_J = 125^\circ\text{C}$		0.60	-	V
	SS515FSH $I_F = 5.0\text{A}, T_J = 125^\circ\text{C}$		0.67	0.77	V
	SS520FSH $I_F = 2.5\text{A}, T_J = 25^\circ\text{C}$		0.78	-	V
	SS520FSH $I_F = 5.0\text{A}, T_J = 25^\circ\text{C}$		0.84	0.95	V
	SS520FSH $I_F = 2.5\text{A}, T_J = 125^\circ\text{C}$		0.64	-	V
	SS520FSH $I_F = 5.0\text{A}, T_J = 125^\circ\text{C}$		0.70	0.78	V
Reverse current @ rated V_R ⁽²⁾	SS54FSH $T_J = 25^\circ\text{C}$	I_R	-	200	μA
	SS56FSH $T_J = 125^\circ\text{C}$		-	40	mA
	SS510FSH $T_J = 25^\circ\text{C}$		-	10	μA
	SS515FSH $T_J = 125^\circ\text{C}$		-	1	mA
	SS520FSH $T_J = 125^\circ\text{C}$		-	1	mA
Junction capacitance	SS54FSH	C_J	256	-	pF
	SS56FSH		245	-	pF
	SS510FSH $1\text{MHz}, V_R = 4.0\text{V}$		164	-	pF
	SS515FSH		126	-	pF
	SS520FSH		100	-	pF

[SS54FSH – SS520FSH 5A, 40V - 200V Schottky Barrier Surface Mount Rectifier](#)

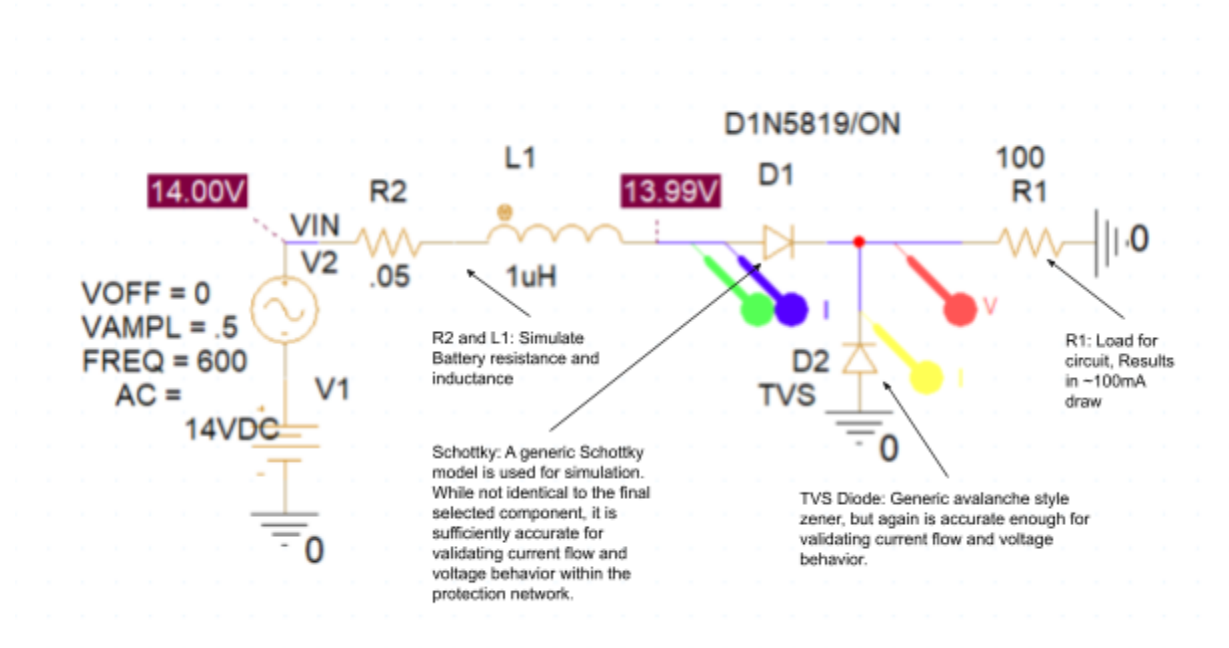
SS54FSH Datasheet

Polarity Testing in Pspice/OrCad Capture

To verify correct operation of the input protection diode, two simulations were performed to demonstrate expected system behavior under both nominal and fault conditions.

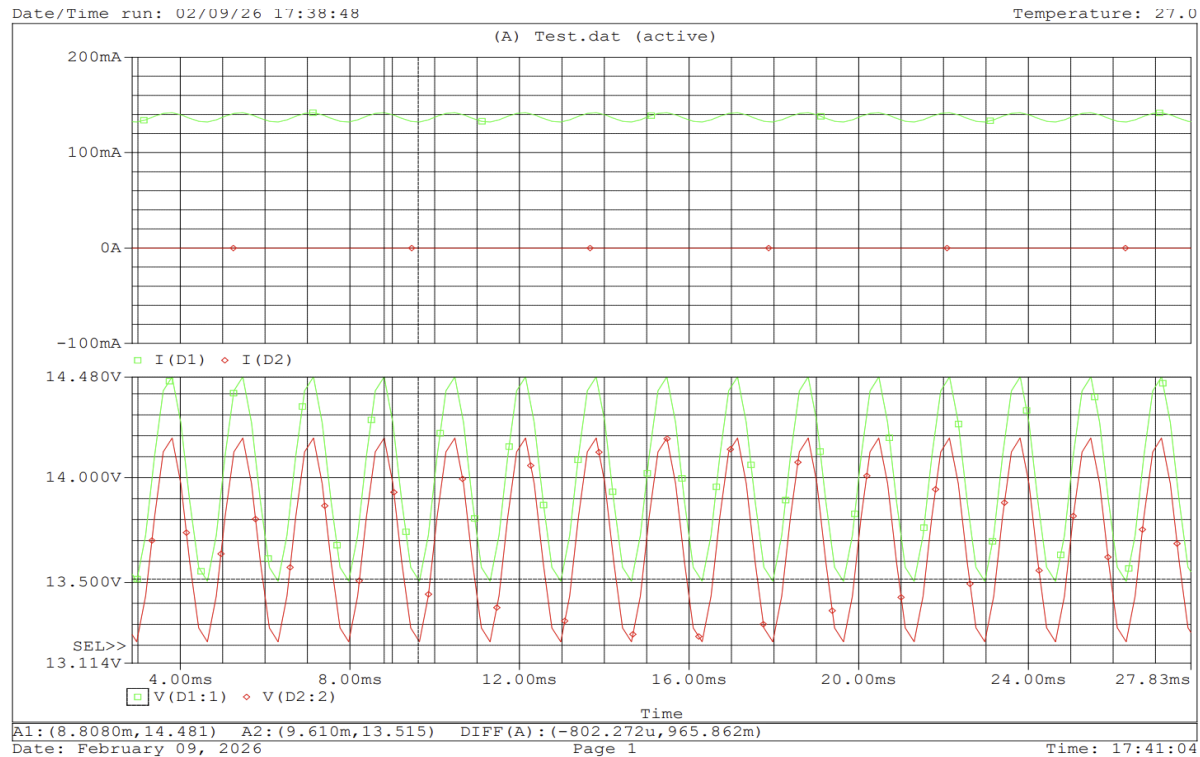
The first simulation establishes a baseline case using a noisy, non-ideal 14 V input supply to represent realistic operating conditions. This test confirms that the diode conducts properly during normal operation and does not introduce unintended voltage drop or instability into the protected rail.

The second simulation evaluates reverse polarity protection. The system is initially operated under normal conditions, after which the input polarity is inverted midway through the simulation. The results confirm that the Schottky diode effectively blocks reverse current, preventing negative voltage from propagating into the protected circuitry.



Normal Operation — Baseline Power Input Behavior

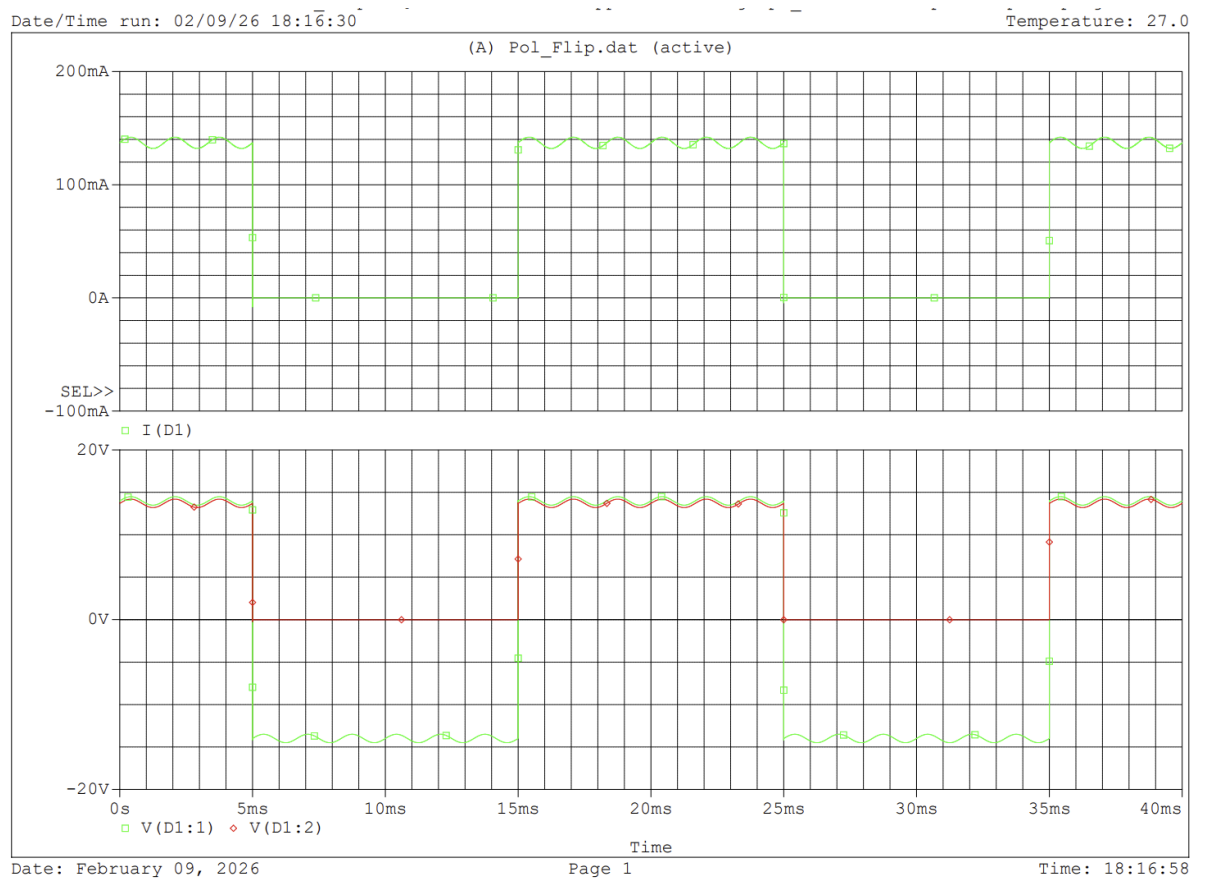
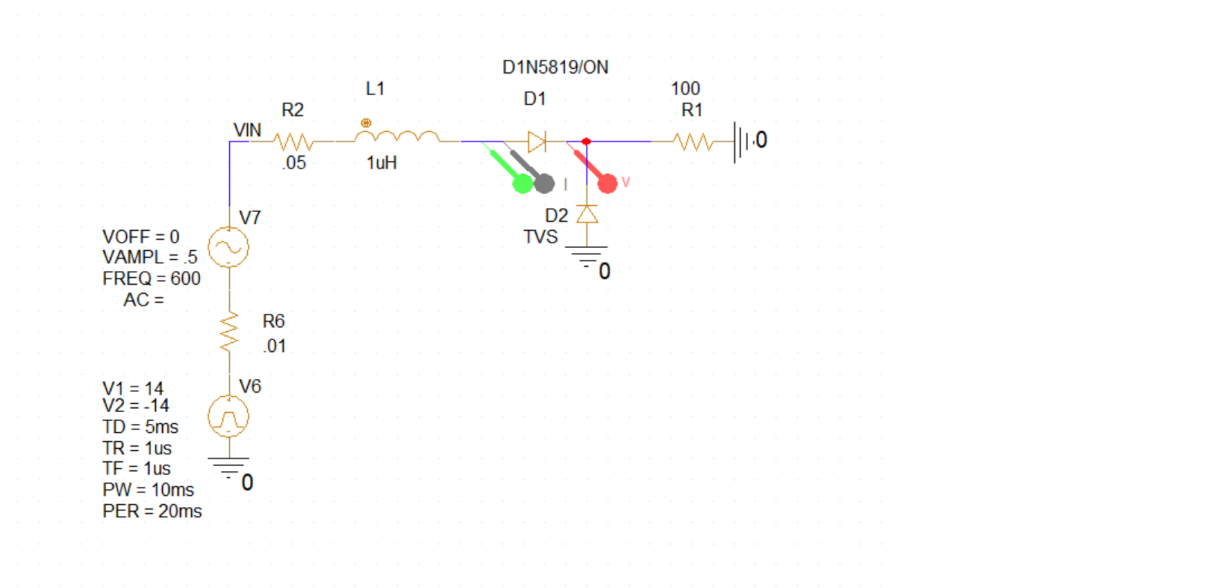
The objective of this simulation is to verify that the input protection network operates transparently under nominal operating conditions. Specifically, the Schottky diode must conduct normally without excessive voltage drop, and the TVS diode must remain non-conductive when the input voltage remains within the normal operating range.



In the normal-operation simulation, the upper plot shows current versus time. The green trace represents the system load current, which in this model is approximately 140 mA. The red trace represents the current through the TVS diode. Under normal operating conditions, the TVS current remains near zero, confirming that the device is not entering breakdown and therefore is correctly sized above the steady-state operating voltage.

The lower plot shows the input voltage before and after the Schottky diode. A small forward-voltage drop is observed across the diode, as expected with non-ideal diodes. However, the resulting downstream voltage remains well above the minimum required input of 7 V for the OKI 5 V regulator. This confirms that the protection diode does not adversely affect normal system operation.

Polarity Flip Test - Schottky Diode Behavior with Lead Reversal



The polarity-reversal simulation is somewhat dense visually due to the superimposed ripple and white background. However, the results clearly confirm correct diode behavior under both nominal and reverse-polarity conditions. In the lower plot, the green trace shows the input node voltage switching from +14 V to -14 V, with an added ripple component to emulate charging-system noise typical of a motorcycle stator/regulator environment. The red trace shows the protected node voltage after the Schottky diode. During the reverse-polarity interval, the protected node does not follow the polarity flip of the input, indicating that the Schottky diode is correctly blocking reverse current and preventing negative voltage from propagating into the downstream circuitry.

The upper plot shows the diode current, which drops to approximately zero during the reverse polarity event. This confirms that the diode transitions from forward conduction to reverse blocking as intended, demonstrating effective reverse-polarity protection for the system.

[AND8220 - How To Use Thermal Data Found in Data Sheets](#)

Report on how to read and use Thermal Data for surface-mount components, and reinforce component selection and copper-pour size for thermal considerations as well.

[Application Note - Basics of Ideal Diodes](#)

Texas Instruments Report on diode applications, including reverse-polarity protection for automotive applications, and explains why a Schottky is present on this board.

SMBJ18A - Transient Voltage Suppression Diode

A TVS diode is used on the 12V input to protect downstream components from electrical transients typical of engine-powered charging systems. On a 2001 SV650, the charging system uses a shunt-style regulator/rectifier, which dissipates excess stator power as heat and runs hot. With 20+ years of use on these R/R systems, the long lifespan increases the likelihood of abnormal conditions, raising the risk of overvoltage events and fast transients at the power input. On top of that, other loads on the motorcycle, such as the starter, relays, and fans, can also generate voltage spikes. The TVS doesn't contribute meaningful heat during regular operation and only conducts during transient events, clamps the input rail to a safe level, and protects the downstream ICs.

The selected TVS, the SMBJ18A, has an 18V Reverse Standoff Voltage, which is above the expected 11-15V standard operating range. The Breakdown Voltage of the SMBJ18A is approximately 20–22 V, at which point the device begins avalanche conduction, while the maximum clamp voltage of approximately 29.2 V remains below the OKI-78SR maximum input rating of 36 V. While the switching regulator has some tolerance to input variation, the TVS provides additional protection by limiting transient overvoltage and reducing electrical stress on downstream components. The maximum clamping voltage is approximately 29.2 V, which limits the peak input voltage during transient events to a level below the downstream regulator's absolute maximum rating. The SV650 permanent-magnet stator can produce up to approximately 70–90 VAC RMS per phase at high engine speeds (and higher under open-circuit conditions). While the regulator/rectifier normally maintains the DC bus near 14 V, aging components, wiring resistance, and transient conditions can allow overvoltage events, justifying the inclusion of input transient suppression.

Unidirectional Device		Bidirectional Device		Breakdown Voltage V _{BR} (Volts)			Working Peak Reverse Voltage	Maximum Reverse Leakage @ V _{RWM}	Maximum Clamping Voltage @ I _{pp} (10/1000 μ s)	Maximum Peak Pulse Current (10/1000 μ s)	Maximum Clamping Voltage @ I _{pp} (8/20 μ s)	Maximum Peak Pulse Current (8/20 μ s)
Part No.	Marking	Part No.	Marking	Min.	Max.	@ I _T (mA)	V _{RWM} (V)	I _R (μ A)	V _c (V)	I _{pp} (A)	V _c (V)	I _{pp} (A)
SMBJ15A	LM	SMBJ15CA	BM	16.7	18.5	1.0	15	1.0	24.4	24.6	31.7	123.0
SMBJ16A	LP	SMBJ16CA	BP	17.8	19.7	1.0	16	1.0	26.0	23.1	33.8	115.5
SMBJ17A	LR	SMBJ17CA	BR	18.9	20.9	1.0	17	1.0	27.6	21.8	35.9	109.0
SMBJ18A	LT	SMBJ18CA	BT	20.0	22.1	1.0	18	1.0	29.2	20.6	38.0	103.0
SMBJ20A	LV	SMBJ20CA	BV	22.2	24.5	1.0	20	1.0	32.4	18.6	42.1	93.0
SMBJ22A	LX	SMBJ22CA	BX	24.4	26.9	1.0	22	1.0	35.5	16.9	46.2	84.5
SMBJ24A	LZ	SMBJ24CA	BZ	26.7	29.5	1.0	24	1.0	38.9	15.5	50.6	77.5
SMBJ26A	ME	SMBJ26CA	CE	28.9	31.9	1.0	26	1.0	42.1	14.3	54.7	71.5
SMBJ28A	MG	SMBJ28CA	CG	31.1	34.4	1.0	28	1.0	45.4	13.3	59.0	66.5
SMBJ30A	MK	SMBJ30CA	CK	33.3	36.8	1.0	30	1.0	48.4	12.4	62.9	62.0
SMBJ33A	MM	SMBJ33CA	CM	36.7	40.6	1.0	33	1.0	53.3	11.3	69.3	56.5

[SMBJ Transient Voltage Suppressor Diode Series](#)

SMBJ18A - Datasheet

OKI-78SR-5/1.5-W36-C (12V > 5V) / TPSM84203EAB (5V > 3V3) Buck Regulators

To maintain stable operation of motherboard-powered modular sensor daughterboards, the distributed supply must remain above the minimum operating voltage of 3.3 V components despite cable resistance, connector losses, and transient load conditions. Therefore, power is distributed at 5 V and locally regulated to 3.3 V at each daughterboard using point-of-load regulation. Distributing power at a higher voltage provides regulation headroom, reducing susceptibility to voltage drop and improving brownout margin and system stability. The 5 V distribution rail also supports future 5 V sensors, improving modularity, expandability, and long-term system flexibility. For these reasons, the board uses two stages of voltage regulation. The primary regulator converts the vehicle's 12 V supply to a stable 5 V rail, which powers the motherboard and serves as the distribution voltage for daughterboards while also allowing USB to power the system during bench testing. A secondary regulator then locally converts 5 V to 3.3 V, supplying the sensitive ICs, including the MCU, CAN transceiver, and other low-voltage components.

With the voltage regulators, input and output capacitors are used to reduce ripple, switching noise, and supply impedance across a wide frequency range. The regulator input uses bulk capacitance comprising two 100 μ F low-ESR capacitors to provide energy storage and reduce low-frequency voltage droop, supplemented by 1 μ F and 0.1 μ F ceramic capacitors placed close to the regulator to improve high-frequency decoupling and mitigate switching noise.

The 5V regulator output uses a 47 μ F ceramic capacitor to reduce output ripple and improve transient response, along with a 0.1 μ F ceramic capacitor placed near the output pins to

suppress high-frequency switching noise and stabilize the local supply rail. The 3V3 output has a .1uF and a 10uF for the same reasons as the 5V output, as well as a 100uF electrolytic cap to help reduce brownouts for the ICs.

https://pim.murata.com/asset/pim4/nonIsolatedDCDCconverter/OKI-78SR-E_PDF_NONISOLATEDDCDCCONVERTER?lastModifiedDatetime=20250707193658

OKI-78SR Data Sheet (Murata)

[TPSM84203, TPSM84205, TPSM84212 1.5-A, 28-V Input, TO-220 Power Module datasheet \(Rev. A\)](#)

TPSM84 Data Sheet (Texas Instruments)